

CLOAKID

By

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SUBJECT: ADSS

DATE: 07/05/2026

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INTRODUCTION

PURPOSE

This Software Requirements Specification (SRS) document describes the CLOAKID system in detail. It describes the system behavior that how will the CLOACKID system will work, its use cases, workflow and features which are used in the system.

The purpose of this document is to provide a clear understanding of the system for developers, instructors and stakeholders. It ensures that all requirements are well defined before the implementation of the system.

DOUCMENT OVERVIEW

This document have several sections which define the full functionality of the system.

INTRODUCTION: It outlines the

- Purpose
- scope,
- key words used in system,
- target audience
- Environment of system
- Constraints
- Assumptions
- Dependencies

Next section will be describing the **distinguished features** of **CLOAKID**. This section will show how it differs from traditional data sharing systems.

Next section describes **USE CASE SPECIFICATIONS**, which shows the interactions between users and the system.

Next, **USE CASE ARTIFACTS**, which define actors, relationships, and system constraints.

The document then includes **USE CASE DIAGRAM** to visually represent system interactions and workflows.

Final section specifies **SYSTEM SCOPE** which defines the boundaries and limitations of the system.

Together, these sections provide a complete and structured description of the CLOAKID platform.

SCOPE

CLOAKID is a secure data sharing platform which is designed to allow users to share sensitive information such as Personally Identifiable Information (PII) and Sensitive PII (SPII) in a controlled manner.

The system enables users to:

- Share data securely with specific individuals verified by the system
- Redaction to hide sensitive portions
- Set access controls such as expiry time and view limits to control unauthorized access
- Monitor data access and activity logs
- Revoke or modify access at any time

A key feature of the system is that the sender has full control over his shared data even after it has been shared.

DEFINATIONS, ACRONYMS AND ABBREVIATIONS

- PII (Personally Identifiable Information): Any data that can identify an individual
- SPII (Sensitive Personally Identifiable Information): Highly sensitive personal data requiring stronger protection
- Sender: User who shares data through the system
- Receiver: User who receives and views shared data
- Encryption: The process of converting data into a secure format to prevent unauthorized access
- Decryption: The process of converting encrypted data back into readable form

- HTTPS (Hypertext Transfer Protocol Secure): A protocol used for secure communication over the internet using encryption (TLS/SSL)
- Redaction: The process of hiding sensitive parts of data (e.g., blur, pixelation)
- Watermark: A visible mark added to data to indicate ownership and prevent misuse

TARGET AUDIENCE

The target audience is:

- Developers and system designers
- Instructors evaluating the project
- Stakeholders interested in system functionality

OPERATING ENVIRONMENT

The CLOAKID system will operate as a web-based application.

It can be accessed through modern web browsers such as Chrome, Firefox, and Edge.

CONSTRAINTS

- System requires continuous internet connectivity
- Only verified users can access and share data
- Data is stored temporarily and deleted after expiry
- Access is limited by time and number of views
- Secure encryption mechanisms must be implemented
- Watermarks will be used to prevent misuse

ASSUMPTIONS

- Users have valid email accounts
- Users have access to modern web browsers

- Email notification service is operational
- Users complete authentication and verification steps

DEPENDENCIES

- System depends on email services for verification and notifications
- System relies on internet connectivity for operation
- Encryption and security mechanisms must function correctly
- Browser compatibility is required for proper usage of the system

DISTINGUISHED FEATURES

1. PROGRESSIVE DISCLOSURE (“REQUEST MORE” REVEAL)

In this feature, the sender does not reveal all the information at once. Instead, only partial and restricted versions of the data are shown to the recipient. If the receiver needs to have closer look he will request more access from the sender, the sender will decide whether to provide access to it or not. Thus, progressive disclosure creates a mechanism of controlled data access.

COMPARISON WITH PREVIOUS SOFTWARE:

Older versions allowed revealing all the data once it was uploaded. It meant that the receiver was able to see everything on the document even the things sender didn't mean to share.

2. SENDER CONTROL AFTER DATA SHARING

This is the key feature of CLOACKID. This feature allows the sender to control his data even after it has been shared. This solution provides maximum control over the data. In this case, the sender has the authority to limit access to the data, block it completely,

adjust visibility options, and apply additional limitations such as viewing time and view counts. As a result, the sender always controls his data.

COMPARISON WITH PREVIOUS SOFTWARE:

Before, users had no way to control the data after data has been shared. Once the file was sent, it could not be taken back or limited further.

3. FOCUSED SOLUTION FOR HANDLING PII AND SPII

As this is a specialized service for working with personal and sensitive data, it offers several unique tools for data processing. These include the options to blur images, pixelate them, and even apply redaction to make sure only the necessary parts of sensitive information are visible to users.

COMPARISON WITH PREVIOUS SOFTWARE:

Traditional software used for handling different types of information was more general. Specific solutions for working with sensitive information were missing from such platforms.

4. STRICT VERIFICATION-BASED SHARING MODEL

Access to this platform for uploading and receiving data is limited to verified email addresses through multi-factor authentication. Both sender and receiver should have verified account to use the platform to share the data. Thus, the system excludes unverified users and share data only between authenticated users.

COMPARISON WITH PREVIOUS SOFTWARE:

Traditional platforms allow the sharing of data through public or semi-public links without verifying who has the link. Anyone with the link can access the data and can re-share it.

In contrast CLOACKID ensure that the data is only shared between verified users and only shared with the person it was intended to be shared with.

5. TIME-BOUND AND VIEW LIMITATIONS

Users may set certain limitations on time and number of viewings allowed to the recipient. Once the deadline has been met, the system disables the recipient access to the data. The system waits for two hours before permanent deletion from the system storage. In this time span the users can regain access to the data. The receiver can request more access time, or the sender can change permissions. After this deletion time, the system removes the data permanently from the storage to ensure integrity and security requirements.

COMPARISION WITH PREVIOUS SOFTWARE:

Data in traditional platforms stayed accessible indefinitely unless it was deleted manually. The data stays in the system which is a threat to confidentiality of the data as it may result in loss of data or data theft. CLOACKID removes the data permanently from the system so that its integrity is maintained and there is no threat for loss.

6. RESEARCHING LIMITED BY DEFAULT WITH POSSIBLE RESHARING + TRACEABILITY

By default, the recipient is unable to forward received information. If the sender decides to allow resharing, it should be restricted to a particular person or a list of pre-approved receivers. Thus, it becomes possible to monitor and control where the information is forwarded to. If the user does not want to add the receiver, it can be shared with anyone having a verified account so that it can be traced.

COMPARISION WITH PREVIOUS SOFTWARE:

Traditional systems do not restrict the resharing of data. It can be forwarded to anyone and there was no trace that who other has the data. This system provides complete logs of whom data has been shared and how many times they accessed that data. This provides the sender with option that if data has been shared with a user who should not see, he can revoke immediate access to that email, and the data will be stopped sharing immediately.

7. PERMENENT WATERMARKS AND ACTIVITY AUDITS

A permanent watermark is added to the information and a log of all actions performed after it has been created. Thus, it is possible to track all data usage and discourage its misuse.

COMPARISION WITH PREVIOUS SOFTWARE:

Watermarking solutions were available before, but tracking and logging actions performed with data were not available before. This system provides complete logs and traceability of the data.

8. LACK OF LONG-TERM DATA STORAGE IN THE EARLY STAGE

Information stored on this platform stays active only until the deletion time. Upon reaching it, the file gets deleted automatically from the storage. This ensures the integrity of the data and reduces the chances of theft.

COMPARISION WITH PREVIOUS SOFTWARE:

Files stayed in traditional solutions forever unless they were deleted manually. This increases the chances of reuse and can become victim of theft.

USE CASES

SENDER USE CASE:

1. REGISTER ACCOUNT

Actor: Sender

Description: Sender creates a new CLOACKID account using his email address.

Preconditions:

- User does not already have an account **Postconditions:**
- Account is created
- Verification email is sent

Flow:

- 1) User enters name, email address, password
- 2) System validates inputs
- 3) Account is created
- 4) Verification email is sent

2. VERIFY EMAIL

Actor: Sender

Description: Confirms identity via email verification link.

Preconditions:

- User has registered
- Verification email has been sent **Postconditions:**
- Account is activated
- User becomes eligible to log in

Flow:

- 1) System sends link
- 2) User clicks link
- 3) Token/OTP is validated
- 4) Account activated

3. LOGIN

Actor: Sender

Description: After email verification and successful creation of the new account user is prompt to enter credentials and login to the system.

Preconditions:

- User account exists
- Email is verified **Postconditions:**
- User session is created
- Access to system is granted **Flow:**

- 1) Enter credentials
- 2) System verifies identity
- 3) Session is created

4. INITIATE DATA SHARING

Actor: Sender

Description: sender starts the sharing process **Preconditions:**

- Sender is logged in **Postconditions:**
- Data sharing process is started

Flow:

- 1) Select “Send Data” option from the dashboard
- 2) Enters receiver email
- 3) System validates receiver and if receiver is a verified user system will show his profile to sender otherwise sharing will be stopped!

5. SELECT SHARING MODE

Actor: Sender

Description: Sender chooses the mode by which data will be shared. Two options will be provided for the sender

- a) Image format
- b) Use templates (pre-built / custom) **Preconditions:**
 - Receiver is verified

- Sender has validated the receiver
- Receiver is verified **Postconditions:**
- Sharing mode (Image/Form) is selected

Flow:

1. Choose Image or Form mode
2. System loads interface

6. UPLOAD / CREATE DATA

Actor: Sender

Description: Sender provides the actual data for sharing. It can be image or form style

Preconditions:

- Sharing mode is selected **Postconditions:**
- Data is uploaded or form is created

Flow:

- 1) Image mode: uploads or capture image from the camera
- 2) Form mode: Uses a template or custom style form and fills the sections with required data for sharing.

7. APPLY REDACTION

Actor: Sender

Description: Hides sensitive portions of data by editing picture if in picture mode.

Preconditions:

- Data is available
- Data is image format **Postconditions:**
- Sensitive areas are hidden or modified **Flow:**

1. Select region / area to hide

2. Apply blur/pixelate/hide
3. Preview changes
4. Specifies the purpose of sharing data
5. Apply watermark (Default purpose)

8. PREVIEW DATA

Actor: Sender

Description: Views final version of data being shared before sending. This will be receiver's view. (What receiver will be able to see will be shown to sender).

Preconditions:

- Data is prepared and redacted **Postconditions:**
- Final view is confirmed by sender

Flow:

1. System renders final view of what is going to be receiver's view
2. Sender verifies content

9. SET ACCESS CONTROLS

Actor: Sender

Description: Defines security rules, restrictions and permissible actions

Preconditions:

- Data is finalized **Postconditions:**
- Access rules (expiry, views, permissions) are defined

Includes:

- Expiry time (12 Hours)
- View limits (Unlimited with allowed time)

- Reshare permission (Default 'NO' allowed with warning to the sender)
- Applied custom water mark (Mandatory)

10. SEND DATA

Actor: Sender

Description: Final transmission of secured data.

Preconditions:

- Access controls are set
- Data is ready for sharing **Postconditions:**
- Data is securely shared
- Receiver is notified **System Actions:**
- Encrypts data using secure protocols
- Generate secure link
- Apply watermark
- Store temporary session

11. MANAGE SHARED DATA

Actor: Sender

Description: User has the control over his shared data after sending **Flow:**

- View active shares
- Monitor access logs

12. MODIFY ACCESS CONTROLS

Actor: Sender

Description: Updates sharing permissions.

Preconditions:

- Data has already been shared **Postconditions:**
- Updated controls or redactions are applied

Options:

- Change expiry time
- Adjust view limits
- Update permissions

13. **REVOKE ACCESS**

Actor: Sender

Description: Sender has option to immediately disable receiver's access at any time

Preconditions:

- Active sharing session exists **Postconditions:**
- Receiver access is immediately blocked **Flow:**

1. Select revoke option
2. System invalidates link and no one can see the shared data

14. **RE-GRANT ACCESS**

Actor: Sender

Description: Restores previously revoked access or grant new permissions **Flow:**

1. Select share session
2. Reactivate permissions
3. Can increase access time or view

15. **TRACK SHARED DATA ACTIVITY**

Actor: Sender

Description: Monitors receiver activity.

Flow:

- View logs
- Check access time
- Track views

16. DELETE SHARED SESSION

Actor: Sender

Description: Permanently removes shared data session. **Flow:**

- Select delete
- System removes all traces

17. RESPOND TO ACCESS REQUEST

Actor: Sender

Description: Handles receiver requests for extra access. **Flow:**

1. Receive request
2. Approve or reject
3. System updates permissions accordingly

18. VIEWS ACCESS AND LOG PERMISSION

Actor: Sender

Description: Sender can view all the logs of access and shared access of his data. He can see who has accessed his data if being re-shared with the time stamp. He will receive the notification for every view.

Flow:

1. Sender shares the data
2. Receiver views it
3. Sender receives notification
4. If sharing is allowed and receiver re-shares data sender gets notified

RECEIVERS USE CASE:

1. LOGIN

Actor: Receiver

Description: Authenticates into system using login credentials

Pre-condition: user has verified account of CLOACKID **Post-condition:** use has logged-in successfully.

Flow:

1. Enter credentials
2. System validates credentials
3. Session is created

2. RECEIVES NOTIFICATION

Actor: Receiver

Description: receiver gets a notification when ever data is shared with him

Precondition: Sender has successfully shared the data. Secure session has been generated by the system.

Postcondition: Receiver has access to the data through dashboard.

Flow:

- 1) Sender shares data
- 2) Session is created and notification is sent to the receiver through push notification and email
- 3) System tells about the access time and other permissions

3. VIEW SHARED DATA

Actor: Receiver

Description: Accesses shared content via secure link or dashboard in app. It allows the receiver to view the shared data within defined access limits

Preconditions:

- Receiver has valid access link or can access it using dashboard •
- Access has not expired

Flow:

1. Open link / dashboard

2. System validates access and verifies permissions
3. Data is decrypted temporarily
4. System checks view limits and expiry time
5. Data displayed securely **Restrictions:**
 - No download allowed
 - No permanent storage
 - Limited by time and views

4. REQUEST MORE ACCESS

Actor: Receiver

Description: Requests extended permissions if receiver thinks the shared data is not enough.

Precondition: Access limit reached or near empty

Postcondition: request is sent to the sender **Flow:**

1. Click request option
2. Send request to sender
3. Await response
4. Gets notification on both acceptance and rejection

5. MARK AREA FOR REVEAL

Actor: Receiver

Description: Requests visibility of hidden portions.

Precondition: Data contains redacted areas

Postconditions: Request is sent to the sender **Flow:**

1. Select hidden region on image or uses permission forms to get more information
2. Submit request

6. REQUEST ACCESS TO MARKED AREA

Actor: Receiver **Description:**

Sends a formal request to reveal specific marked areas if the provided data is not enough

Preconditions:

- Area must be selected or permission form must be

filled **Postconditions:**

- Sender receives request **Flow:**

1. System forwards request
2. Sender reviews
3. Decision is made
4. Notification is sent according to the decision made

7. VIEW UPDATING DATA

Actor: Receiver **Description:**

Receiver views updated data after sender approval.

Preconditions:

- Sender has approved request **Postconditions:**

- Data is updated by the sender and is displayed **Flow:**

1. Receiver opens link again through email or dashboard
2. System validates updated permissions
3. Updated data is shown to the receiver

SYSTEM USE CASE:

1. VERIFY RECEIVER IDENTITY

Actor: System

Description: system ensures that the receiver is a verified user of the CLOACKID before showing the receiver's profile to the sender for validation.

Preconditions:

- Receiver email is provided

Postconditions:

- Access is granted or denied **Flow:**
 1. System checks receiver email
 2. Matches with stored data
 3. Checks whether receiver has verified email or not
 4. Grants or denies access

2. CREATE SECURE COPY (TEMPORARY)

Actor: System Create Secure Copy (Temporary)

Description: the actual data is not shared with any user. System makes a temporary copy of the data for sharing.

Preconditions:

- Sender initiates the sharing of data **Postconditions:**
 - Temporary secure copy is created

Flow:

1. System duplicates data
2. Stores it in temporary secure memory for sharing

3. ENCRYPT DATA

Actor: System

Description: Encrypts sensitive data before temporary storage and transmission to ensure confidentiality. Data transmission is secured using HTTPS(TLS).

Preconditions:

- Data is ready for transmission
- Data is securely encrypted

Flow:

1. Apply encryption algorithm
2. Generate encrypted version of the data
3. Store encrypted version in temporary secure storage
4. Transmit data securely over HTTPS(TLS)

4. APPLY WATERMARKS

Actor: System

Description: Adds watermark for security and traceability. This watermark will be customized by the sender or by default uses the purpose of sending data along with string “NOT VALID FOR OTHER PURPOSES” and will be attached to the data.

Preconditions:

- Data copy exists
- User has provided either ‘purpose’ of sharing or customized

watermark

- Watermark embedded

Flow:

1. Generate watermark from the purpose of sharing data or uses the customized watermark provided by the sender
2. Embed into data

5. GENERATE SESSION ON THE DASHBOARD OF RECEIVER AND SENDS NOTIFICATIONS

Actor: System

Description: Creates a session on receivers profile/dashboard and sends notifications through email and app push notifications.

Preconditions:

- Data is encrypted and prepared **Postconditions:**
- Secure session is created
- Notification sent to the receiver

Flow:

1. Generate secure session on receiver's profile
2. Send notifications

6. DECRYPT DATA (AT VIEW TIME)

Actor: System

Description: Decrypts data when the receiver accesses it.

Preconditions:

- Valid access request **Postconditions:**
- Data is temporarily decrypted **Flow:**
 1. Validate request
 2. Decrypt temporarily
 3. Displays data until permissible

7. ENFORCE VIEW LIMIT

Actor: System

Description: Ensures that every view reduces the view count and notification is sent to the sender for every view. When view count equals '0' access is revoked.

Preconditions:

- View limit is defined by the sender, or system uses default in access time

Postconditions:

- Access allowed or denied based on count **Flow:**
 1. Track each access
 2. Compare with '0'
 3. Allow or deny

8. ENFORCE EXPIRY TIME

Actor: System

Description: Ensures that after the time limit has reached the access is restricted immediately.

Preconditions:

- Expiry time is set by the sender **Postconditions:**
- Access granted or blocked based on time **Flow:**
 1. Check current time
 2. Compare with expiry time
 3. Allow or block

9. AUTO DELATE DATA

Actor: System

Description: Systems waits for 2 hours for sender or receiver to make changes to access time or views or regain access. After 2 hours of expiration time the data is permanently removed from the system along with its original copy. (If users want to regain access, they will need to restart the process after deletion time).

Preconditions:

- Expiry conditions are met
- Users didn't regain access

- Deletion deadline of '2' hours is met **Postconditions:**
 - Data permanently deleted **Flow:**
1. Detect expiry condition
 2. Waits for two hours for users' action
 3. Remove data permanently

10. Track Activity & Audit Logs

Actor: System

Description: Records all system actions. From sending to receiving, if the data has been re-shared (if allowed), their email addresses and view count. For all these things sender is notified.

Preconditions:

- Any system/user activity occurs

Postconditions:

- Activity is recorded securely **Flow:**
1. Log user activity
 2. Store logs securely

11. Send Notifications

Actor: System

Description: Notifies sender about receiver's actions, viewing, sharing or requesting more information or access.

Preconditions:

- Trigger events occur (view/request/re-shares)

Postconditions:

- Notification sent to sender for every access **Flow:**
1. Detect event (view/request)

2. Send notification

USE CASE ARTIFACTS

1. ACTORS

Sender (Primary Actor)

The sender is the main user of the system who initiates data transmission in a secure environment. The sender has full authority over his shared data even after it is sent.

Responsibilities:

- Register account
- login into the system using verified credentials
- Initiate data sharing
- Enter and verify receiver details
- Choose sharing mode (Image/Form)
- Upload or create data
- Apply redaction (blur, pixelate, hide)
- Preview final data
- Add watermark
- Set access controls (expiry time, view limits, reshare permissions)
- Send data securely
- Monitor receiver activity
- Modify, revoke, or re-grant access upon requests

Receiver (Secondary Actor)

The receiver is the user who receives and views shared data under restricted conditions defined by the sender.

Responsibilities:

- Receive notification of shared data
- Access data using secure link
- View only permitted content
- Check remaining views and time
- Request more views or time

- Mark hidden portions for reveal
- View updated data after approval

System (Internal Actor)

The system is responsible for enforcing all security, processing, and control mechanisms. It is responsible for strictly using file transfer protocols for secure transfer of data.

Responsibilities:

- Verify receiver identity
- Encrypt and decrypt data
- Generate secure links
- Apply watermark to data (Default or user provided)
- Enforce access rules (expiry, views, no re-shares (by default))
- Track user activity (logs)
- Send notifications to both sender and receiver
- Automatically delete data after deletion time

Admin

The admin supervises system operations and ensures smooth functioning. He receives feedback from the users and use it for improvement of the system and increase user trust.

Responsibilities:

- Monitor system activity
- Manage users if required
- Maintain system integrity
- Receive feedback
- Improve system defects

USE CASE RELATIONSHIPS

- The **Sender** is connected to all data creation, control, and management use cases
- The **Receiver** interacts with viewing and request-related use cases

- The **System** supports all use cases by performing internal operations such as validation, encryption, and enforcement of rules defined by primary actor
- Sender and Receiver can be considered specialized roles of a general **User**

SYSTEM BOUNDARY

INSIDE THE CLOAKID SYSTEM

- Data encryption and decryption
- Secure session/token generation
- Temporary data storage during session deletion time (no long-term storage)
- Watermarking mechanism
- Access control enforcement (views, expiry, share permissions)
- Activity logging and monitoring
- Notification handling

OUTSIDE THE CLOAKID SYSTEM

- Sender (Primary)
- Receiver (Secondary)
- Admin **Explanation:**

All sensitive operations are handled within the system boundary. Users interact with the system through define interfaces, ensuring secure and controlled access to the data. The system is responsible for enforcing the permission given by the sender. The system does not store the data permanently for security reasons. The data will be completely removed from the system storage after deletion deadline meets.

CONSTRAINTS:

- Data is stored temporarily and deleted after deletion time
- Access is limited by time and number of views
- Only verified users can send or receive data
- Receivers cannot download or permanently store data
- Resharing is disabled by default

- Watermark is required either customized by the sender or by default purpose of sharing data will be used as watermark
- All user activities are logged
- System requires internet connectivity

ASSUMPTIONS:

- Users have valid email addresses
- Email address belongs to the user creating account
- Users have access to internet and modern browsers
- Email notification system is functional
- Users complete verification before using the system

BUSINESS RULE:

- Sender always retains full control over shared data
- Receivers can only access data within defined permissions by sender
- Access can be modified or revoked by sender anytime
- Data access is controlled by expiry time and view limits
- After expiry, data becomes inaccessible until regaining access is requested or sender changes time (with two hours of expiry time)
- After deletion time (two hours of expiry), data is permanently removed from the system
- Any request to reveal hidden or redacted data must be approved by the sender
- All actions are recorded for accountability

The diagram illustrates the CLOAKID SYSTEM architecture, divided into three main sections: SENDER USE CASE, CLOAKID SYSTEM INTERNAL USE CASES, and RECEIVER USE CASE.

SENDER USE CASE: This section includes use cases for the sender actor, such as Register, Login, Verify Email, Initiate data sharing, Enter Receiver Email, Confirm Receiver Identity, Select Sharing Mode, Update Data, Apply redaction, Preview Data, Add Watermark, Set Access Controls, Send Data, Manage access (modify, inspect, revoke access), and View access logs.

CLOAKID SYSTEM INTERNAL USE CASES: This section includes internal system use cases, such as System check receiver is verified, Show receiver profile to sender, Create receiver copy (temp. no storage), Encrypt data end-to-end, SS. Apply watermark, Generate secure session (precondition for notification), Decrypt data view time, Set view limits (view time, expiry time), Track activity & audit log, and Send notification to receiver.

RECEIVER USE CASE: This section includes use cases for the receiver actor, such as Receive notification, View shared data, Check remaining views & time, Request more views & time, Mark area for reveal, Request access to marked area, and View update data.

The diagram also includes a section for ADMIN/OPTIONAL use cases, which includes A1. LOGIN, A2. MONITOR SYSTEM ACTIVITY, and A3. MANAGE USERS.

Relationships between use cases are indicated by solid lines (actors to use cases) and dashed lines (use cases to use cases). Some relationships are labeled with stereotypes like «include», «extend», and «precondition».

This use case diagram shows the interaction between sender, receiver and System. It shows how at every level the user interacts with the system. This diagram illustrates how authentication, encryption, and access limitations regulate secure data exchange. It also shown that the system internal operations include management of verification, monitoring, and controlled access.

SCOPE OF THE SYSTEM

(CloackID)

CloackID is a secure data-sharing platform that is used to send sensitive information safely from one person to another person. The main purpose of this platform is to prevent the personal information of the person from unauthorized access and being misused.

This system mainly focuses on protecting PII (Personal Identifiable Information) and SPII (Sensitive Personal Identifiable Information). This includes

- ID numbers
- Contact details
- Sensitive documents
- Medical records
- Criminal records
- Domicile information

And many more information that should not be exposed publicly or cannot afford misuse.

In CloackID, only verified users are allowed to use the platform. Every user must register his account going through the authentication process. It includes multifactor authentication. This ensure that the person whom the data is being shared is a verified and actual person.

This system includes secure encryption of the data both at rest and during transit. This means that the data is encrypted and shared over HTTPS which uses TLS to protect data during transmission. This ensures that if someone tries to intercept it they cannot read the data or misuse it.

OBJECTIVES OF THE SYSTEM:

The objectives of CLOAKID are:

- To provide a secure platform for sharing sensitive information
- To ensure that only verified users can send and receive data
- To provide sender control even after data has been shared

- To reduce the risk of unauthorized access and data misuse
- To maintain complete activity logs and traceability
- To protect sensitive information using encryption and access controls

WHAT CLOAKID COVERS:

- The scope of CLOAKID includes:
- User registration and account verification
- Multi-factor authentication
- Secure sharing of PII and SPII
- Image-based and form-based data sharing
- Redaction, blur and pixelation of sensitive information
- Secure transmission of data using HTTPS (TLS)
- Data encryption before storage and transmission
- Sender controlled access permissions
- Expiry time and view limitations
- Access revocation and re-granting
- Progressive disclosure of information through access requests
- Watermarking of shared information
- Activity logs and audit trails
- Notifications for user activities
- Temporary storage of shared information
- Automatic deletion of data after expiry conditions are met.

Target users:

The target users of the system are:

- Sender: User who shares sensitive information
- Receiver: User who receives and views shared information
- Admin: User responsible for monitoring system operations and maintaining system integrity

System Boundary:

The CLOAKID system is responsible for:

- Verification of users
- Encryption and decryption of data
- Secure sharing of information
- Access control management
- Watermark generation
- Activity logging and monitoring
- Notification handling
- Temporary storage and deletion of data

The system is not responsible for external internet services, third-party email providers or user devices.

WHAT CLOAKID DOES NOT COVER:

The system does not cover:

- Public file sharing
- Anonymous users
- Social media communication
- General purpose cloud storage
- Long-term storage of user data
- Real-time messaging services
- Collaborative document editing
- Sharing with unverified users

Future Scope:

In future versions, the system can be extended to include:

- Mobile application support
- Biometric authentication
- AI-based sensitive data detection
- Advanced audit reporting
- Enterprise level access management
- Integration with government identity verification systems
- Real-time messaging service

The CLOAKID system is designed to provide a secure and controlled environment for sharing sensitive information while ensuring that the sender remains in control of the data throughout its lifecycle. The system emphasizes privacy, confidentiality, integrity and accountability in every stage of the data sharing process.

CONCLUSION

The CloackID is a system that is made to share sensitive data with other users in a secure and traceable environment. It emphasizes securing the data via encryption, authentication, and access controls system. This system helps stop illegal use and data breaches by making sure that data is shared only with verified and intended users. It provides full control to the user over his data. A sender can set access time, view limits, and can revoke access at any time. With default no share option and traceability logs, this platform enhances privacy. This system enhances privacy, security, integrity, and confidentiality in online data transmission while keeping complete sender control over their data.